

# Financial News Sentiment - Corrected Comparison Report

Dataset note: Hugging Face currently loads 9,543 train rows and 2,388 validation rows. The proposal/dataset-card text lists 9,938/2,486, so the code now warns about this mismatch and uses the official loaded splits.

Corrected validation results: Simple RNN accuracy 72.03%, macro F1 0.6322. LSTM accuracy 78.27%, macro F1 0.7093. GRU accuracy 79.10%, macro F1 0.7117. BERT/FinBERT accuracy 82.83%, macro F1 0.7598.

Best baseline: GRU. BERT leads the corrected GRU by +3.73 percentage points accuracy and +6.76% macro F1.

Root cause 1: RNN/LSTM/GRU used the final padded timestep. The models now use the last real token representation and were retrained.

Root cause 2: saved BERT label metadata used negative/neutral/positive. The config and loaders now use 0=Bearish, 1=Bullish, 2=Neutral.